GaMPF: A Full-Scale Gated Message Passing Framework Based on Collaborative Estimation for VHR Remote Sensing Image Change Detection

Xiaoyang Zhao, Keyun Zhao, Siyao Li, Chuanming Song, and Xianghai Wang

***Abstract*— With the maturity and popularization of high-performance sensor technology, it is now possible to acquire huge amounts of very high-resolution (VHR) remote sensing (RS) images. The change detection (CD) for VHR images is currently receiving special attention for RS Earth observation applications; however, as a hot research field, it needs to be studied in depth to improve the detection accuracy of fine changes. To this end, a full-scale gated message passing framework (GaMPF) based on collaborative estimation (CE) for VHR RS image CD is proposed in this article. On the one hand, the key embedding representation is generated for each feature map by means of the CE strategy. On the other hand, grounded in timing analysis, bitemporal features are sent selectively on dual paths according to the full-scale gated (FsG) mechanism. Specifically, this framework consists of the following four components: 1) taking shared-weights Siamese network as an encoder to extract multiscale features; 2) generate a set of shared compact bases under the CE strategy and infer the key embedding representations on the basis of the shared bases for feature maps at the same level, considering the representations as the gated switches; 3) FsG mechanism is used as the mode of message passing between bitemporal images, which guides the information can be transmitted simultaneously on both within- and cross-temporal paths; and 4) creating a stepwise dense fusion module (DFM) as a decoder for predicting the change map. The experimental results show that the GaMPF proposed in this article outperforms existing state-of-the-art (SOTA) methods and is particularly good at detecting edges and small objects. The source code will be released at https://github.com/zxylnnu/GaMPF.**

***Index Terms*— Change detection (CD), convolutional neural network, full-scale fusion, gated function, very high-resolution (VHR) remote sensing (RS) image.**

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I. Introduction

REMOTE sensing (RS) image change detection (CD) technology plays a particularly important role in Earth observation applications. In the early days, due to the limitations of imaging equipment, most RS images taken were medium and low resolutions, and the phenomenon of mixed elements prevailed, which was not conducive to capturing small changing objects. In recent years, with the improvement of Earth observation technology with the characteristics of "high precision, all-round, short period" and the development of optical sensors, very high-resolution (VHR) images have become a new carrier of RS targets, and at the same time, VHR images also provide the possibility to obtain fine change information of land features, which is thus urgently needed by many research fields such as map making, disaster assessment, as well as urban planning [1].

The precise geometric and positional information contained in VHR images can clearly reflect the ground object features of various scales in the observed area, making it possible to capture multiscale changed features. However, the diverse land cover environment in practical applications creates complex VHR imaging information, which undoubtedly brings new challenges for VHR-CD [2]. The challenges specifically reflect in: first, the diversity of ground object expression, which is mainly manifested in the multiple scales, various shapes, and random positions of features; second, the heterogeneity of changed ground object expression, including multiple interferences caused by seasonal changes, shadow coverages, and other factors; and third, the sensitivity of multitemporal VHR images to time periods, which greatly increases the difficulty of controlling pseudo-change information [3]. Accordingly, how to ensure correct prediction of changed feature positions while capturing the changed features as accurately as possible remains an important and difficult problem to be solved in this field.

In early studies, unsupervised algebraic and transformation methods are basic methods to solve this problem due to the low resolution and limited number of processed images, such as iteratively reweighted multivariate alteration detection [4], change vector analysis [5], and principal component analysis [6]. The main idea of these methods is to project image data

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to the transform domain and then maximize the difference between changed and unchanged features by thresholding or clustering methods. Threshold-based methods are sensitive to subjectively selected thresholds and have empirical errors, while cluster-based methods are not robust enough to outliers. With the significant increase in spatial resolution of sensors, VHR images present rich and diverse features, enabling the development of some object-based CD methods [7]. Such methods improve the integrity and continuity of detected edges by exploiting contextual neighbor information. However, the detection accuracy usually relies on hand-crafted classifiers and ignores the influence of another temporal image.

In recent research, benefiting from a large number of visual interpretation work and the improvement of hardware computing capabilities, deep-learning-based methods have become a current research hotspot in the field of CD by virtue of powerful nonlinear data fitting and automated processing capabilities [8]. Many effective frameworks have been proposed in recent years. Papadomanolaki et al. [9] use fully convolutional long short-term memory (LSTM) networks to model spatial–temporal joint feature representation; Saha et al. [10] use graph convolutional networks to capture spatial and spectral features; Fang et al. [11] integrate "aggregation-distribution" and "feature exchange" feature interaction strategies based on the transformer framework to fuse the bitemporal features; He et al. [12] leverage the effectiveness of feature pyramid networks in bridging the semantic gap and use a cross-resolution attention module to aggregate contextual information cross-resolutions. These frameworks have different strengths and weaknesses; for instance, although the method based on the LSTM framework can model temporal correlation, the input patch is usually small, limiting the perception of spatial context, while the method based on graph neural networks can alleviate the need for large-scale training data, but the detection accuracy relies on super-pixel segmentation; transformer-based methods require slicing as well to overcome the heavy computational burden associated with multihead self-attention.

As one of the most popular CD frameworks nowadays, the Siamese network uses two shared weight branches to perform feature extraction from bitemporal input samples and output their deep feature representation, which can stand out the similarities and differences between bitemporal features. Codegoni et al. [13] introduce EfficientNet64 as the backbone to extract multilevel features from bitemporal images and design a lightweight feature mixing strategy with feature-level concatenation, group convolution, and multilevel perceptron. Lee et al. [14] take SE-ResNet50 as the feature extractor and propose a local similarity attention module to enhance change information. Pan et al. [15] propose a densely connected feature fusion network to compensate for the high-frequency information loss caused by continuous upsampling and downsampling in the encoder, and combined convolutional block attention module (CBAM) to enhance the deep representation of features at each level. To sum up, although the existing methods have achieved good detection results, many issues need to be further considered, which are analyzed as follows.

First of all, according to previous experience, the use of multiscale features is one of the most significant means to achieve accurate detection of multiscale change entities [16]. This type of method can be summarized as the "multiscale fusion" method (see Section II-A for details). It compares the bitemporal feature maps extracted by the backbone network at each scale and then decodes them in depth to obtain the final change map. These methods explore spatial–temporal correlations profoundly from the multiscale perspective, but it is obvious that such fusion mode introduces some invalid and pseudo-change information, which may dilute real change information. Second, the existing thinking on the attention mechanisms also has certain limitations (Section II-B). The attention mechanisms used in convolutional neural network (CNN)-based VHR-CD methods usually appear in two forms: one is dual attention module (DANet [17]) class, which obtains the correlation between any two positions in space or channel in the form of dense encoding; the other is CBAM [18] class, which obtains strong response information from both spatial and channel dimensions by pooling operations. Both of them generally act on the fused features to enhance the representation of change information, but this approach undoubtedly neglects the preservation of high-response features at each scale. If they are applied to multiscale feature maps, a heavy computational load will be brought. Finally, according to the fact that the larger the receptive field, the more information the network perceives and the better the detection results [19], the multiscale context modeling method also needs to be refined and studied in depth to further enhance the network detection performance.

In view of the above problems, a full-scale gated message passing framework (GaMPF) based on collaborative estimation (CE) for VHR RS image CD is proposed. The main contributions are given as follows.

1) A novel framework with the full-scale gated (FsG) mechanism is proposed, which is different from the existing three mainstream feature fusion frameworks. It realizes full-scale within- and cross-temporal messages passing simultaneously, and uses the gated function to reduce the receiving of similar information within-temporal path and increase the sending of change information cross-temporal path.
2) The CE strategy is proposed to generate a key embedding for each pair of bitemporal feature maps and regard the embeddings as gated switches. It enables coupling learning of channel and spatial attention based on a set of compact bases.
3) To reduce the gap between shallow pixel-level detail information and deep semantic-level change information, the dense fusion module (DFM) is proposed as the decoder to predict the change map in a stepwise manner, so that the fused features have both global and local information.

The rest of this article is organized as follows. Section II presents some related work referred to this article, including the structure of CNN-based VHR-CD networks and the use of attention modules by each method. Section III introduces the proposed GaMPF in detail. Section IV provides the experiment

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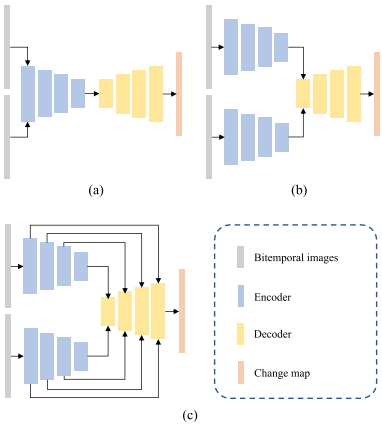


Fig. 1. Three popular CNN-based VHR-CD network structures. (a) Single scale-level early fusion. (b) Single scale-level late fusion. (c) Multiscale-level fusion.

and analysis, including experimental setup, datasets, and other related content, as well as the presentation of comparison and ablation experiments. Section V concludes the whole article.

II. Related Works

This section will review the CNN-based VHR-CD methods in terms of both the network structure and the use of attention mechanisms.

A. CNN-Based VHR-CD Network Structures

Benefiting from the high-performance feature transmission design of U-Net, pixel-level prediction tasks can be learned in an end-to-end mode. This framework, summarized as "encoder–decoder," has been utilized in a variety of deep learning-based image processing fields such as image restoration, semantic segmentation, and style transfer. As shown in Fig. 1, the structure of CNN-based VHR-CD networks can be broadly classified into three categories according to the location of feature fusion, namely, "single-scale prefusion (early stage)," "single-scale postfusion (late stage)," and "multiscale fusion (intermediate stage)."

1) Single-Scale Prefusion: Take the U-Net framework as the main body, bitemporal images are first concatenated or subtracted, pretrained models such as Resnet and VGG are used as the backbone to obtain multiscale features, and finally decode it to the original size step by step to predict the change map. For instance, Chen et al. [20] take the concatenated bitemporal image as the input and to obtain more accurate detection edges by proposing a new training strategy with edge loss; Song and Jiang [21] integrate spatial attention module and channel attention-guided interference filtering unit with atrous spatial pyramid pooling to highlight the difference between changed entities and the background in the concatenated image; Chouhan et al. [22] introduce the concatenation and difference of bitemporal images into the network, and reinforce the change information through symmetrical downsampling and deconvolution operations; and Song et al. [23] take advantage of CNN and transformer backbone to extract features from the stacked images, and afterward, axial attention and cross attention are employed to enhance feature representation from different dimensions.

2) Single-Scale Postfusion: Due to the sequential nature of the CD task, the Siamese encoder–decoder network structure with dual inputs has become one of the most popular network structures in this field. Different from the prefusion-like network, such networks encode multiscale spatial features in parallel from shallow to deep, using shared weights backbones. After that, capture the difference between the fused deep features and finally project the difference to the change map. There are some representative works. Chen and Shi [24] use the self-attention-based pyramid spatiotemporal attention model to obtain multiscale spatiotemporal representation after integrating bitemporal image features extracted by Resnet into the same scale; Fang et al. [25] combine the Siamese network with NestedUNet organically as SNUNet-CD, and the most representative features are extracted by ensemble channel attention module; Chen et al. [26] apply the spatial and channel dual attention modules to the extracted deep features, using L2 distance to measure the difference between features; and Zhang and Shi [27] transfer the pretrained Sub-VGG16 network for image classification to extract depth information and obtain change information through differential operations.

3) Multiscale Fusion: It can be seen that the single-scale fusion modes described above are mostly either concatenating or difference operations on the shallowest or deepest features. However, it is well-known that to generate a high-level embedding representation for each pixel, the size of the feature map is successively reduced and increased, where shallow feature maps contain a large amount of low-level visual information such as edges, textures, small objects, and displaying spatial features, which are capable of accurately acquiring boundaries; as the number of layers in the network deepens, shallow features are gradually abstracted into deep features, which express some high-level information about semantic, contextual, and other relationships, and the dilution of shallow features results in a loss of accurate edge information. As a result, the former of the network structure loses temporal information, and the latter loses fine-grained spatial information. For this reason, some works have adopted the multiscale fusion mode to obtain valid information in feature maps at each scale. Specifically, they model the temporal correlation of feature maps at each scale extracted by the backbone, regarding as joint "spatial–temporal" multidimensional information, and obtain the change map through a decoder. Lei et al. [16] present a multilevel fusion network with a paired channel fusion module, which integrates change features captured by the reverse spatial attention mechanism from coarse to fine. Chen et al. [28] propose EGDE-Net for focusing on boundary accuracy guarantee and change region integrity, capturing multiscale difference information stepwise to get accurate change object boundaries. Ke and Zhang [29] enhance the information flow between different scale feature maps with cross and hierarchical residual connections, and on this basis, capture long-range relationships from three dimensions of height, width, and channel. Zhang et al. [30] enable the model to generate more accurate edges by proposing a feature difference module, which highlights

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change information in the difference map from low to high level.

Furthermore, a number of networks combining the above-mentioned fusion ways are proposed. As proof, Lei et al. [31] combine "single-scale prefusion" and "multiscale fusion," and design a multilevel feature network that extracts the difference between bitemporal images to improve the discriminative power of the network on telling the difference between change targets and background. In addition, methods combined "single-scale postfusion" and "multiscale fusion" have been developed, in which Lei et al. [32] employ a spatial–spectral nonlocal module with multiscale parallel sampling to enhance the internal consistency of change entities and refine the change information through the differential enhancement module at all levels. Zhao et al. [33] use the spatial and semantic feature aggregation module to integrate high-level and low-level spatial information and calculate the difference to strengthen the change information. Chen et al. [34] utilize the multilevel feature encoding capability of U-Net, and they concatenate the feature maps of all levels obtained by the attention mechanism step by step and then project the change features from deep to shallow.

It can be seen that the network structure of CNN-based VHR-CD has already provided some research on the fusion patterns between cross-temporal feature maps, but there is still a lack of difference analysis between cross-temporal feature maps with the same scale, as well as correlation exploration among the same temporal feature maps with different scales.

B. Use of Attention Mechanisms in VHR-CD

Mining strong response information in feature maps is important for obtaining change information. Among them, the attention mechanism is widely used in the VHR-CD task by assigning higher weights to changed features so that the network can focus on specific information while reducing the interference of irrelevant information. The two most commonly used attention modules in CNN-based VHR-CD networks are the CBAM and DANet. The former is expanded on the basis of SE-Net [35], and the latter combines the nonlocal module instantiated by the self-attentive mechanism [36] with CBAM.

CBAM can enhance feature representation in both spatial and channel dimensions in turn, which has a high lightness, strong portability, good effectiveness, and other advantages. In more detail, first of all, its channel attention module and spatial attention module both consist of some pooling, mapping, and simple calculation operations, avoiding the computational load caused by large-scale matrix operations or convolutional operations, and is efficient to train; afterward, it can be integrated into an end-to-end network in a plug-and-play manner for training, and is easy to implement; and finally, it analyses the features in multiple dimensions, showing not only the regions that play an effective role in the spatial domain but also the importance of each channel, jointly improving the network performance.

The nonlocal module obtains a long-range dependence by calculating the correlation degree of any two positions in the feature map. It first calculates the correlation intensity matrix of two positions, the softmax normalization operation is followed to get the attention map, and the features are weighted according to the map to make similar features strongly correlated. This self-attention-based global relationship exploration mode mainly contributes in two aspects: on the one hand, it makes up for the disadvantages such as computational inefficiency and optimization difficulties of the original reliance on stacked convolutional layers to expand the receptive field; on the other hand, it breaks through the long-range feature capture capability that is not available due to the image-slicing operation in modeling spatiotemporal correlation [37].

In addition, the DANet, which can adaptively integrate local features with global dependencies, is proposed, aiming to acquire semantic interactions in spatial and channel dimensions in a parallel manner [38]. Specifically, it first uses the original nonlocal module to implement the position attention mechanism, that is, to selectively aggregate the features of other positions through the weighted sum of all position features in the space, and promote the expression of similar features; subsequently, a channel attention module with a similar principle as position attention module is developed, which enhances the role of high-response channels in the network by mining the potential relationship between any two channels; and lastly, the outputs of the two modules are summed to further enhance the feature representation, but CNN-based VHR-CD networks applying the self-attention mechanism often have a certain computational load [24].

III. Methodology

This section describes the proposed GaMPF structure in detail: first, the overall process is provided in Section III-A; after that, the designed novel CE strategy, FsG mechanism, and DFM are explained in Sections III-B–III-D, respectively; and finally, the loss function is given in Section III-E.

A. Network Overview

The overall architecture of GaMPF is shown in Fig. 2, which consists of the following four parts.

1) A weight-sharing Siamese network structure is used as the backbone feature extractor, and the input bitemporal VHR images are defined as T1,T2 $\in \mathcal{R}^{H\times W\times C}$, where H, W, and C denote the height, width, and number of channels, respectively, and the multiscale features $\mathrm{Ft}_m^n (n = 1, 2; m = 1, 2, 3, 4)$ are extracted by Resnet18 [39], where $n = 1$ denotes the previous temporal image, $n = 2$ denotes the temporal image, and m represents the number of feature scales, the larger the scale, the deeper the features are.
2) According to the CE strategy introduced in Section III-B, key embedding representations are generated for the cross-temporal same-scale feature maps. It receives Ft_m^1 and Ft_m^2 feature maps as input and generates spatial descriptors att_m^n for each temporal feature map based on an estimated set of shared

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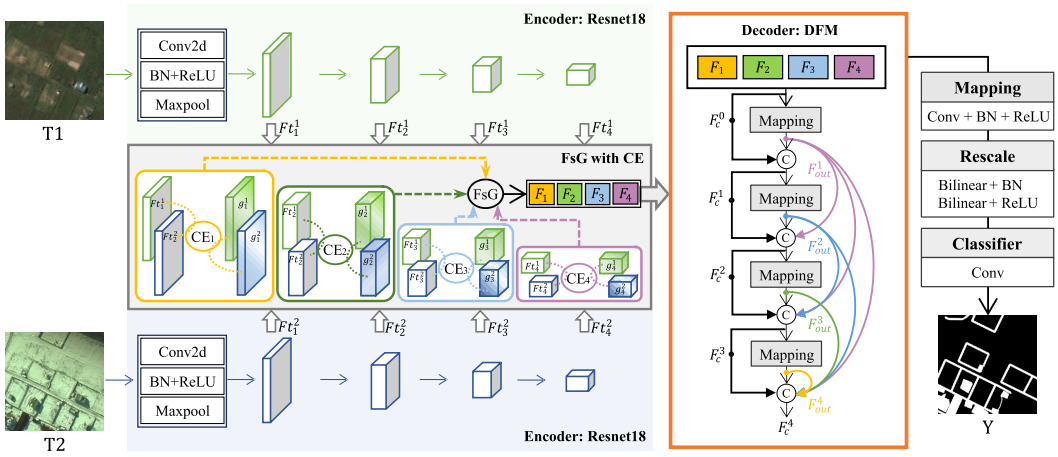


Fig. 2. Illustration of the overall GaMPF architecture.

compact bases γ_m. Then, multiply them together as switches g_m^n of the gated function.

3) Embed CE into FsG, the input of the FsG mechanism with CE is Ft_m^n, g_m^n, and acquiring the spatiotemporal joint feature F_m according to message transfer pattern between bitemporal images designed in Section III-C, so that the full-scale features can be sent on the path of within- and cross-temporal simultaneously, and under the supervision of the gated function, most of the sent information is valid, reducing the interference of irrelevant information.

4) To obtain more comprehensive context information, the decoder DFM receives the concatenated F_m as input (denoted as F_{FsG}). It fuses the features from deep to shallow in a gradually connected way, ensuring that the network can retain deep semantic-level change information without losing the shallow pixel-level detail information, and the output of DFM is F_c^4. In the end, the spatial size of F_c^4 is restored to the size of input images through the mapping layer, rescale layer, and classifier layer. The output of the network is denoted as Pred. The process of GaMPF is summarized in Algorithm 1.

Algorithm 1 Flowchart of GaMPF

Input: A pair of bitemporal VHR RSIs, $\{(T1^i, T2^i)\}_{i=1}^N$; Ground truth, $\{Y^i\}_{i=1}^N$; Learning rate, lr; Training epochs, E; Optimizer $Adam$;

Output: The prediction change map,$\{Pred^i\}_{i=1}^N$;

for epoch $\in [1, E]$ **do**
  #Step1: Multi-scale feature extraction by Resnet18
  $\{Ft_m^1\}^i = \{[Ft_1^1, \ldots, Ft_4^1]\}^i = Resnet18(\{T1\}^i)$;
  $\{Ft_m^2\}^i = \{[Ft_1^2, \ldots, Ft_4^2]\}^i = Resnet18(\{T2\}^i)$;
  #Step2: Obtain switches through gated function with CE strategy
  $\{g_m^1, g_m^2\}^i = CE(\{Ft_m^1, Ft_m^2\}^i)$;
  #Step3: Propagate the information by FsG mechanism
  $\{F_m\}^i = FsG(\{Ft_1^1, \ldots, Ft_4^2, g_1^1, \ldots, g_4^2\}^i)$;
  #Step4: Fuse pixel- and semantic-level features through DFM decoder
  $\{F_{FsG}\}^i = concat(\{F_1\}^i, \{F_2\}^i, \{F_3\}^i, \{F_4\}^i)$;
  $\{F_c^4\}^i = DFM(\{F_{FsG}\}^i)$;
  #Step5: Obtain the change map
  $\{Pred\}^i = Classifier(Rescale(Mapping(\{F_c^4\}^i)))$
  #Step6: Calculate the loss
  $\{Loss_{BCE}\}^i = L_{BCE}(\{Pred\}^i, \{Y\}^i)$
  # Step7: Employ $Adam$ with lr to optimize
end for

B. CE Strategy

Enhancing information representation from spatial and channel multiple dimensions is currently one of the most effective means of capturing change information. The independent learning mode between the spatial and channel attention modules makes the information they generate individually of each other, and there is also some redundancy in channelwise feature descriptors. In contrast, the proposed CE strategy requires to be constructed based on a set of compact bases $\mu_m^n (n = 1, 2; m = 1, 2, 3, 4)$ representing channel descriptors, so that the channel descriptors can be more representative, and take advantage of these bases, the corresponding spatial descriptors att_m^n are obtained, and project att_m^n to the input feature map to update μ_m^n; subsequently, the shared channel descriptor γ_m is extracted, and the product of it and each temporal spatial descriptor is the key embedding representation of each feature map; furthermore, it is known as the switch of the gated function as well. Such a design makes the shared representations in the same scale feature maps to be consistent and, meanwhile, highlights the representations of different features. The specific flow of the CE strategy is shown in Fig. 3, and the concrete steps are given.

1) Feature Transformation: CE strategy receives $\mathrm{Ft}_m^1, \mathrm{Ft}_m^2 \in \mathcal{R}^{(H/2^{m+1})\times(W/2^{m+1})\times 64m} (m = 1, 2, 3, 4)$ extracted by Resnet18 as input. As well-known that, when too many channels exist, there will be a certain similarity among channels, which will cause redundancy and increase the computational cost as well. Therefore, the transformation layer with convolution and Sigmoid is used first to project the input to a low-dimensional feature map, with 64 channels. For convenience of expression

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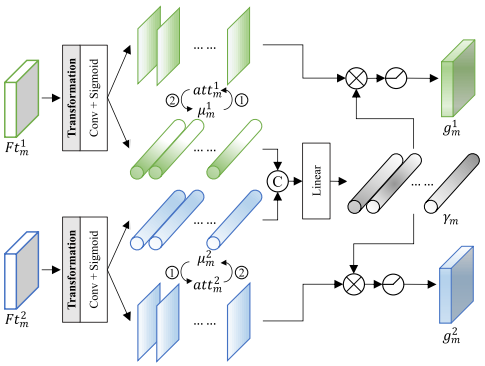


Fig. 3. Flowchart of CE strategy.

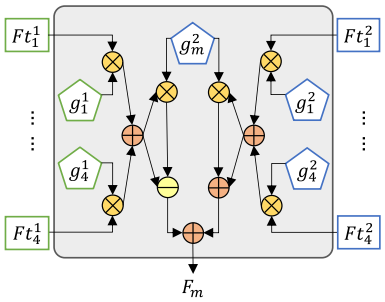


Fig. 4. Flowchart of FsG mechanism.

and calculation, reshape Ft_m^1 and Ft_m^2 as $hw \times c$, where $hw = (H/2^{m+1}) \times (W/2^{m+1})$.

2) Obtain Each Temporal Channel Descriptor: First, initialize a set of bases $\mu_m^n \in \mathcal{R}^{k\times c}$ to compactly express the input. Since the difference between shallow features is stronger than that of deep features, defining $k = (64/2^{m-1})$, as the scales are larger, the value of k decreases. When $k = x$ and $c = y$, its physical meaning is the weight of the xth base to the yth channel. And then, the $\mathrm{att}_m^n \in \mathcal{R}^{hw\times k}$ corresponding to μ_m^n is calculated according to the following formula. When $hw = x$ and $k = y$, its physical meaning is the weight of the yth base to the xth position in Ft_m^n

$$\mathrm{att}_m^n(hw,k) = \mathrm{softmax}\left(\mathrm{Ft}_m^n(hw,c) \times \left(\mu_m^n(k,c)\right)^{\mathrm{T}}\right). \quad (1)$$

Next, update μ_m^n according to the following formula:

$$\mu_m^n(k,c) = \left(\frac{\mathrm{att}_m^n(hw,k)}{\sum_{hw=1}^{HW}\mathrm{att}_m^n(hw,k)}\right)^{\mathrm{T}} \mathrm{Ft}_m^n(hw,c) \quad (2)$$

where HW is the product of the height and width of att_m^n.

3) Obtain Gated Switches: With the aim of extracting some common features γ_m from bitemporal images, the shared channel descriptor generation process shown in the following equations is designed, and on this basis, the response of each spatial descriptor is combined to finally obtain the key embedding representation g_m^n of each feature map, which is used as a switch for message passing to realize both capturing the invariant information between images by the shared channel descriptor and reinforcing the change information by combining each spatial descriptor:

$$\gamma_m = \mathrm{Linear}\left(\mathrm{concat}\left(\mu_m^1, \mu_m^2\right)\right) \quad (3)$$

$$\begin{cases} g_m^1 = \mathrm{ReLU}\left(\mathrm{att}_m^1 \times \gamma_m\right) \\ g_m^2 = \mathrm{ReLU}\left(\mathrm{att}_m^2 \times \gamma_m\right). \end{cases} \quad (4)$$

C. FsG Mechanism

For the multitemporal analysis task of CD, the importance of modeling temporal correlation has been self-evident. How to filter out pseudo-change information from a large amount of confounding information while preserving real-change information is a key point in problem-solving and also a difficult problem. In this article, an FsG mechanism is designed to enable efficient message passing on within- and cross-temporal paths simultaneously.

Fig. 4 shows the propagation process of the FsG mechanism for the mth scale, which receives the multiscale features Ft_m^n extracted by Resnet18 and $g_m^n (n = 1, 2; m = 1, 2, 3, 4)$ estimated by the CE strategy as input. It can be observed that the FsG starts with unification Ft_m^n and g_m^n to the same size as Ft_1^1 and g_1^1, after that, the output F_m can be acquired through some simple calculations. The following formula symbolizes and deduces the FsG. The process of FsG can be divided into the following steps. To begin with, the message of each feature map is generated in a weighted summation manner, and g_m^n and Ft_m^n are regarded as the weight coefficient and the feature map, respectively; afterward, g_m^2 is used as the gated switch of each temporal total message after aggregation to control whether and how much to send messages to Ft_m^2 feature map; and lastly, the learning content of the module is optimized by the residual learning strategy

$$\begin{aligned} F_m &= \left(1 + g_m^2\right)\sum_{m=1}^{4} g_m^2 \mathrm{Ft}_m^2 + \left(1 - g_m^2\right)\sum_{m=1}^{4} g_m^1 \mathrm{Ft}_m^1 \\ &= g_m^2\left(\sum_{m=1}^{4}\left(g_m^2\mathrm{Ft}_m^2 - g_m^1\mathrm{Ft}_m^1\right)\right) + \sum_{m=1}^{4}\left(g_m^2\mathrm{Ft}_m^2 + g_m^1\mathrm{Ft}_m^1\right). \end{aligned} \quad (5)$$

Through the above process, FsG realize full-scale message passing, with equipped gated switches that suppress similar message passing within the same temporal path while intensifying valid messages passing across the other temporal path to avoid message redundancy and can reinforce the propagation of mutually exclusive messages from other scales to adjust the edges of change targets.

D. Dense Fusion Module

The necessity of modeling multiscale context information has been confirmed [12]. In order to obtain more valid information in the feature map, a novel DFM is designed as a decoder, as shown in the orange box in Fig. 2. The DFM takes the concatenation of F_1, F_2, F_3, and F_4 as input, which are the multiscale output feature maps of the FsG mechanism. In detail, the DFM is a combination of mapping layers, concatenation operations, as well as short- and long-skip connections, where the mapping layer consists of convolution, batch normalization (BN), and rectified linear unit (ReLU)

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layers. The outputs of the four mapping layers shown in Fig. 2 are named F^s_{out}, C means concatenation operation, and the features after concatenation are recorded as F^s_c. The joint representation of the DFM is as follows.

If $s = 0$

$$F^s_c = F_{\text{FsG}} = \text{concat}(F_1, F_2, F_3, F_4) \tag{6}$$

else if $s = 1, 2, 3, 4$

$$\begin{cases} F^s_{\text{out}} = \text{Mapping}\left(F^{s-1}_c\right) \\ F^s_c = \text{concat}\left(F^{s-1}_c, F^s_{\text{out}}, \ldots, F^1_{\text{out}}\right). \end{cases} \tag{7}$$

The above DFM can integrate deep-level features into shallow-level features multiple times to achieve the purpose of introducing rich semantic information into pixel-level feature maps, so that the low-level features can also perceive high-level context information, making the fused feature map with both overall and local detail information, thus enabling the network to have stronger feature representation capability.

E. Loss Function

For the purpose of training the end-to-end GaMPF, the vanilla binary cross-entropy (BCE) loss function shown in the following is used to calculate the loss between the predicted change map Pred and the ground true Y:

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{x,y=0}^{N}\begin{pmatrix} Y_{x,y}\log \text{Pred}_{x,y} \\ +\left(1 - Y_{x,y}\right) log\left(1 - \text{Pred}_{x,y}\right) \end{pmatrix} \tag{8}$$

where N is the number of elements in Y; $\text{Pred}_{x,y}$ is the coordinate value at position (x, y) in the predicted change map, which represents the probability that the network predicts the element will change; and $Y_{x,y}$ is the value of the ground truth at the same position, showing the change category, and there are two possible values, where 0 means that the point has not changed, and 1 means that the point has changed.

IV. Experiment and Analysis

This section gives specific details of the experiments, including some subjective and objective comparison results with state-of-the-art (SOTA) methods, ablation experiments, as well as efficiency analysis.

A. Datasets and Experimental Setup

To verify the effectiveness of GaMPF, a series of experiments is conducted on three publicly available high-resolution change detection dataset (CDD) [40], cropland change detection (CLCD) dataset [41], and satellite side-looking (S2Looking) dataset [42]. CDD is one of the most commonly used datasets in this field, while CLCD is a recently proposed dataset that has not been widely used, and S2Looking is a large-scale RS building CDD. Some test samples of the three datasets are shown in Fig. 5, and the details are given in the following.

1) *Seasonal-Varying CDD*: Taken by Google Earth, with spatial resolutions ranging from 3 to 100 cm/pixel, it consists of 16 000 pairs of RS images with a size of 256 × 256 × 3 after cropping, of which 10 000 pairs are used for training, 3000 pairs for validating, and 3000 pairs for testing. It is a large-scale binary CDD with multiscale change targets, such as cars, buildings, roads, and so on, and has obvious interference factors due to seasonal changes.
2) *CLCD*: Collected by Gaofen-2 in Guangdong, China, in 2017 and 2019, proposed in response to the current lack of detection targets for farmland changes. It consists of 600 pairs of 512×512×3 farmland RS images, of which 360 pairs are used for training, 120 pairs for validating, and 120 pairs for testing. The spatial resolution is distributed in the range of 0.5–2 m/pixel. The main change targets include buildings, lakes, and lands.
3) *S2Looking*: Captured by GaoFen, SuperView, and BeiJing-2 satellites from 2017 to 2020. It contains 5000 registered bitemporal image pairs with a size of 1024 × 1024 × 3. Different from other datasets, S2Looking contains large-scale side-looking satellite images captured at varying off-nadir angles, which makes it challenging. Taking the computational load into account, we crop the dataset into 256 × 256 × 3 patches and obtain the training, validation, and testing datasets consisting of 56 000, 8000, and 16 000 patches, respectively.

It can be seen that the detection of all datasets requires multiscale feature analysis, the shape as well as the size of the change targets is also extremely diverse, which greatly increases the detection difficulty, and the complex background also affects the robustness of the network.

In order to compare the detection results of different methods more objectively, six commonly used objective evaluation metrics are used to comprehensively measure the detection performance, namely, precision (P), recall (R), F1-score ($F1$), overall accuracy (OA), Kappa (K), and intersection over union (IoU). The definitions are given as follows:

$$\begin{cases} P = \dfrac{\text{TP}}{\text{TP} + \text{FP}}, \quad R = \dfrac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = \dfrac{2P \times R}{P + R} \\ \text{OA} = \dfrac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad K = \dfrac{\text{OA} - \text{Pre}}{1 - \text{Pre}} \\ \text{IoU} = \dfrac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \end{cases} \tag{9}$$

where TP, TN, FP, and FN represent the true positives, true negatives, false positives, and false negatives, respectively. The values of the above six metrics range from 0 to 1, and the higher the value, the more correct the detection result is. And the intermediate value Pre is calculated as follows:

$$\text{Pre} = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{TN} + \text{FP})(\text{TN} + \text{FN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{10}$$

This experiment is implemented under the PyTorch framework, and the training and testing are completed on an Intel Xeon workstation with one Nvidia Titan X (Pascal) GPU. The Adam optimizer [43] is used for training, and the learning rate is set to 0.001 with $\beta 1 = 0.9$ and $\beta 2 = 0.999$, where $\beta 1$ and $\beta 2$ are exponential decay rates for the moment estimates. After fixing 150 epochs, the training is terminated. The batch size is

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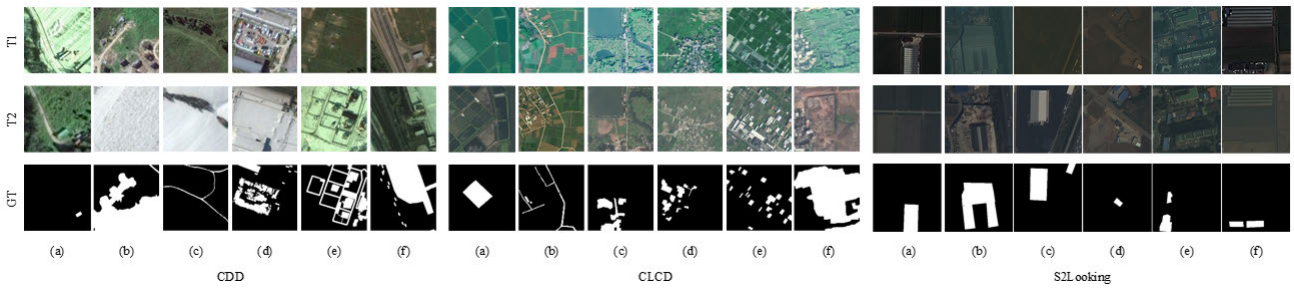


Fig. 5. Test samples of CDD, CLCD dataset, and S2Looking dataset. (a) Sample 1. (b) Sample 2. (c) Sample 3. (d) Sample 4. (e) Sample 5. (f) Sample 6.

TABLE I
OBJECTIVE METRICS OF EACH METHOD DETECTION RESULTS FOR THE CDD

	Transformer-based methods					CNN-based methods					
	SwinSUNet 2022	SARAS-Net 2022	RCDT-SwinT 2022	STDecoder-CD 2022	SiamixFormer 2023	SSESN 2022	SMD-Net 2022	FCCDN 2022	SDACD 2022	DSAHRNet 2023	GaMPF
P	95.7	97.76	96.63	94.39	-	97.3	96.6	97.33	97.13	96.38	**98.75**(↑**0.99**)
R	92.3	97.23	96.97	89.72	-	96.2	97.4	93.93	97.56	97.51	**98.33**(↑**0.77**)
$F1$	94.0	97.49	96.80	92.00	97.13	96.7	97.0	95.60	97.34	96.94	**98.54**(↑**1.05**)
OA	98.5	99.35	-	-	-	-	99.3	-	-	99.27	**99.64**(↑**0.29**)
K	-	-	-	-	-	-	96.6	-	-	-	**98.33**(↑**1.73**)
IoU	-	95.11	93.79	85.17	94.51	-	94.2	91.57	-	94.07	**97.12**(↑**2.01**)

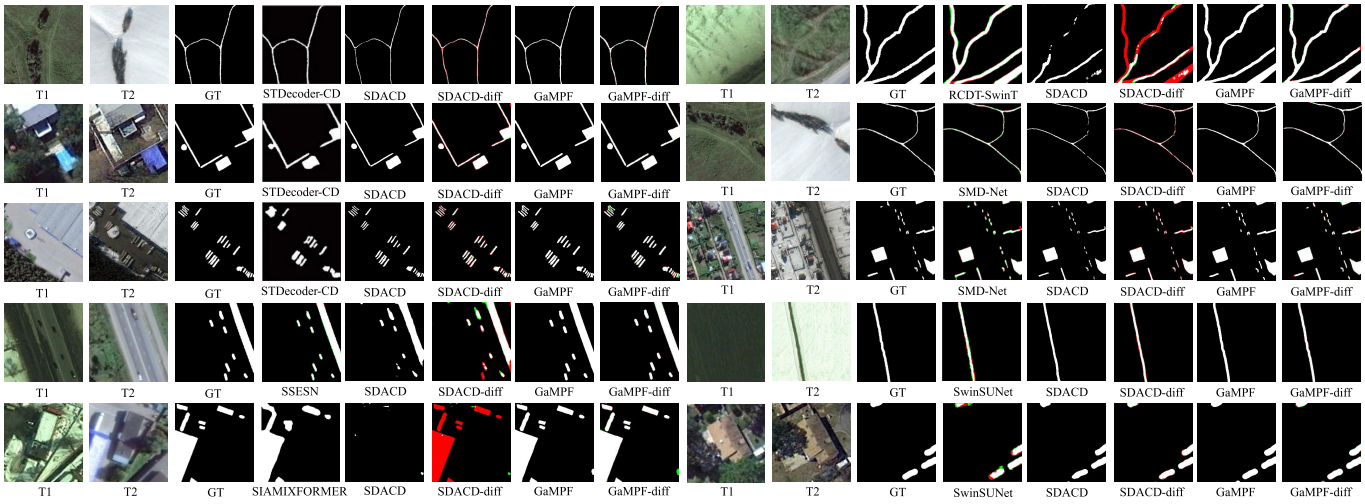


Fig. 6. Detection results of each comparison algorithm on the test samples with small change targets in the CDD.

set to 16 for training the CDD, 4 for training CLCD dataset, and 16 for S2Looking dataset. Vertical flips, horizontal flips, and 90°, 180°, as well as 270° rotations are implemented to augment the training data of CDD and CLCD dataset.

B. Comparison Experiments on CDD

To illustrate the excellent detection performance of the GaMPF proposed in this article on the CDD, we compare a series of recently published excellent methods and divide them into two categories. One is transformer-based: SwinSUNet [44], SARAS-Net [45], RCDT-SwinT [46], STDecoder-CD [47], and SiamixFormer [48]; and the other is CNN-based: SSESN [33], SMD-Net [30], FCCDN [49], SDACD [50], and DSAHRNet [51]. The objective metrics of each method detection results for the CDD are given in Table I. The experimental data and predicted change maps of all comparison methods come from each originally proposed article. The best result of each metric is marked in bold, the suboptimal result is underlined, and the growing value of the method in this article over the best result in the comparison algorithms is provided.

It can be observed from Table I that the detection results of the proposed GaMPF for CDD have achieved the highest values in all objective metrics, the F1 and IoU values are 1.05% and 2.01% higher than the suboptimal comparison methods, respectively. Moreover, the difference between P and R is also relatively small, indicating that the detection performance of the proposed method is relatively stable, and it can achieve high accuracy while guaranteeing superior recall. Meanwhile, the higher K value also shows a high degree of consistency with ground truth (GT).

Figs. 6 and 7 present the detection results and difference maps of some comparison methods listed in Table I for testing

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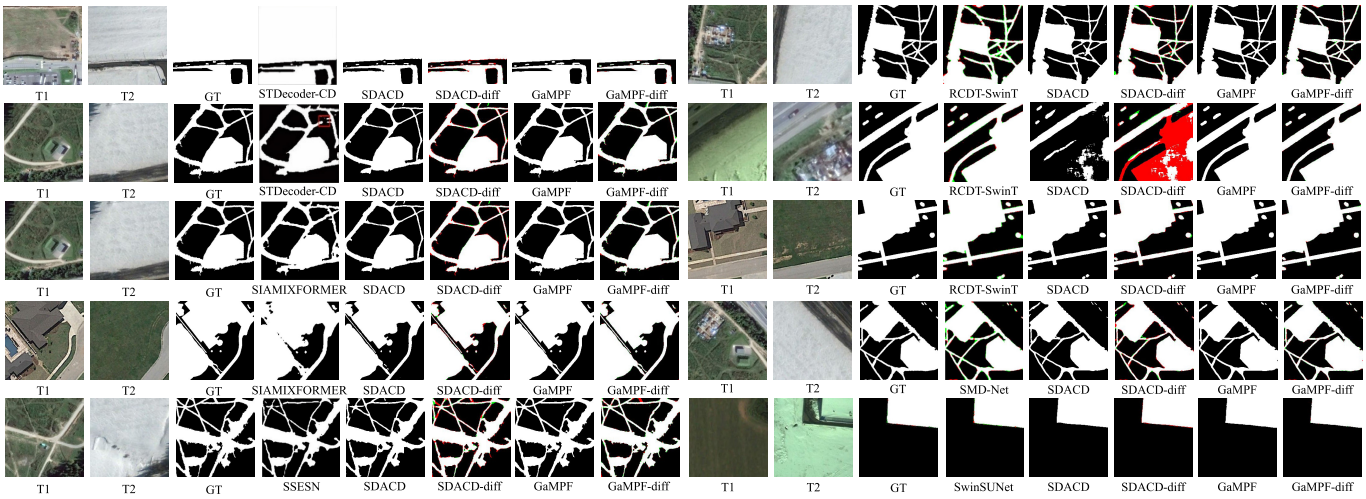


Fig. 7. Detection results of each comparison algorithm on the test samples with large change targets in the CDD.

TABLE II
OBJECTIVE METRICS OF EACH METHOD DETECTION RESULTS FOR THE CLCD DATASET

	DTCDSCN 2021	BiT 2022	MFPNet 2021	MSCANet 2022	DSAMNet 2022	SNUNet-CD 2022	DR-TANet 2021	GaMPF
P	54.49	61.42	83.20	75.36	82.16	79.44	80.52	**84.06**(↑0.86)
R	66.23	62.75	60.74	67.64	62.88	64.30	65.90	**68.36**(↑0.72)
$F1$	59.79	62.08	70.22	71.29	71.24	71.07	72.48	**75.40**(↑2.92)
OA	-	-	-	-	96.22	96.11	96.28	**96.68**(↑0.40)
K	-	-	-	-	69.26	69.01	70.51	**73.64**(↑3.13)
IoU	42.64	45.01	54.11	55.39	55.33	55.13	56.84	**60.52**(↑3.68)

the samples in the CDD. The detection results are all from the original proposed articles. Moreover, in difference maps, the green pixels are false alarms, and the red pixels are missed alarms, to show the differences with GT maps more visually.

Fig. 6 shows the test samples with small change targets in the CDD. It is easy to observe that the detection results of GaMPF outperform those of SDACD. First, GaMPF can capture changed objects more accurately with lower missed alarms, and second, it is more consistent with GT at the edge of detecting changed objects. Especially for the prediction of roads and cars, it is not only possible to maintain the continuity of the road, but also detect the scattered cars. In addition, the detection results of STDecoder-CD and SIAMIXFORMER shown in Fig. 6 are coarse, and they are weaken at detecting small changed entities. Furthermore, the difference maps of SMD-Net and SwinSUNet contain a lot of green pixels, demonstrating that there are many false negatives, and the detection of these two methods tends to be conservative. In contrast, the proposed GaMPF shows the optimal detection performance, with much less red and green pixels in each difference map than those of other methods.

Fig. 7 shows the test samples with large change targets in the CDD. As can be seen from the left side of Fig. 7, the STDecoder-CD and SSESN methods are not precise enough to detect the details, and the detection of bar objects is often not continuous. In addition, there are many false alarms in the detection results of SIAMIFORMER, and many unchanged objects are lost. Furthermore, there are often many red pixels in the difference maps of SDACD detection results, indicating that the method may lack the ability to fully capture the changed information. As can be seen from the right side of Fig. 7, there are more green pixels in the differential diagram of RCDTSwinT and SMD-Net, and most of them occur at the edge of the changed objects, indicating that the two methods are not robust enough to identity pseudo-change information at edges.

C. Comparison Experiments on CLCD Dataset

In order to present the excellent detection performance of GaMPF on the CLCD dataset, Table II shows the objective detection results of existing known methods on the CLCD dataset. As shown in Table II, the comparison methods are: BiT [52], MSCANet [41], DTCDSCN [38], MFPNet [53], DR-TANet [54], DSAMNet [55], and SNUNet-CD [25]. The best detection result in each metric is marked in bold, the suboptimal result is underlined, and the growing value is provided as well.

The objective metrics provided in Table II show the superior detection performance of GaMPF on detecting the CLCD dataset, with all the metrics obtain the highest values. Specifically, the P and R values are 0.86% and 0.72% higher than the suboptimal results, resulting in a significant increase of 2.92% F1 value and 3.68% IoU value. DTCDSCN and BiT both obtain F1 values less than 70 and IoU values less than 50, illustrating that the capability of these two methods

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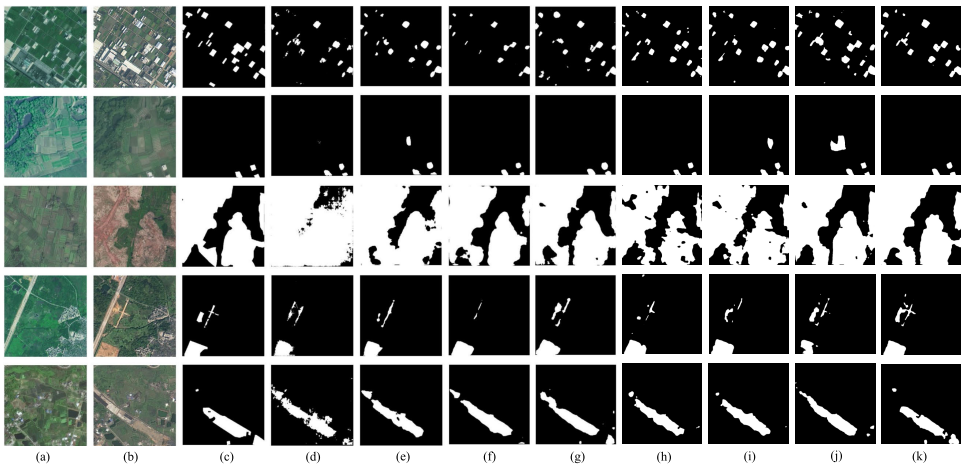


Fig. 8. Visualization of experimental results on the CLCD dataset. (a) T1. (b) T2. (c) GT. (d) DTCDSCN. (e) BiT. (f) MFPNet. (g) MSCANet. (h) DSAMNet. (i) SNUNet-CD. (j) DR-TANet. (k) GaMPF.

on capturing real change information is poor, and fail to effectively identify the change regions. Moreover, the P values of DSAMNet and MFPNet are relatively high, especially for MFPNet, which achieves the suboptimal P value, but due to the great difference from the R values, $F1$ values of these two methods are affected badly. It indicates that the $F1$ metric, as the harmonic mean of P and R, can only become high when the difference between P and R is small and both of them are high. In contrast, MSCANet obtains the suboptimal R value, and a relatively small gap between the P and R, thus obtaining a higher F1 value. Furthermore, DSAMNet and SNUNet-CD obtain similar F1 and IoU values, while SNUNet-CD obtains a higher P value and DSAMNet obtains a higher R value, indicating that SNUNet-CD can predict more changed pixels, while DSAMNet predicts more conservatively.

Fig. 8 shows the detection results of some test samples in CLCD dataset. It can be observed that these methods can coarsely locate to the changed area coarsely; however, the detection result maps of DTCDSCN look messy, and some of them are invalid. The rest of the methods have varying degrees of false and missed alarms. By contrast, the background in the detection result map of the method proposed in this article is cleaner, and the detection of small targets is also more accurate, not only represented in terms of position to locate the changed region, but also the edges of the detected targets.

D. Comparison Experiments on S2Looking Dataset

In order to further verify the effectiveness of GaMPF on large-scale datasets, a comparison experiment with DSAMNet on the S2Looking dataset is conducted. Table III provides a detailed objective comparison with all six indicators. It can be observed that the detection level of GaMPF is superior to that of DSAMNet. Fig. 9 shows the visualization comparison of experimental results on the S2Looking dataset.

The difference maps in the former two lines of Fig. 9 illustrate the effectiveness of GaMPF in capturing changed features compared with DSAMNet, the middle two lines show the superiority of GaMPF at detecting edges, and the last two lines show that GaMPF not only captures the changed features, but also suppresses the pseudo-changed features.

TABLE III
OBJECTIVE METRICS OF EACH METHOD DETECTION RESULTS FOR THE S2LOOKING DATASET

Year	Method	P	R	$F1$	OA	K	IoU
2022	DSAMNet	69.82	46.87	56.09	99.10	55.65	38.96
	GaMPF	73.46	48.68	58.56	99.17	58.15	41.40

E. Ablation Study of GaMPF

In order to verify the importance of each part in GaMPF, essential ablation experiments are carried out in this section. According to the relevant content above, there are three parts need to be ablated. First of all, it is necessary to clarify the network structure of the baseline, which uses the simplest difference between bitemporal feature maps at the same scale and combined with bilinear interpolation to replace the mentioned parts; afterward, to explore the impact of CE, w/o CE experiment is defined as the ablation experiment of CE, which uses convolution followed by Sigmoid function to replace CE; then, FsG ablation experiment is defined as w/o FsG, which replaces FsG with a simple subtraction operation; and finally, the effect of DFM is verified through w/o DFM ablation experiment. Table IV gives the objective detection results of GaMPF ablation experiments.

Table IV shows that each ablation experiment has a certain improvement over the baseline. Fig. 10 provides visualization comparison result of the ablation experiments on three datasets in the form of difference maps, where the green pixels marked are false alarms and the red pixels marked are missed alarms, furthermore, the total number (colored black) of misclassified pixels and the corresponding IoU values (colored blue) are given below each difference map. It can be seen that the proposed GaMPF can capture the real changed information as much as possible, and, at the same time, suppress the influence of the pseudo-changed information on the network.

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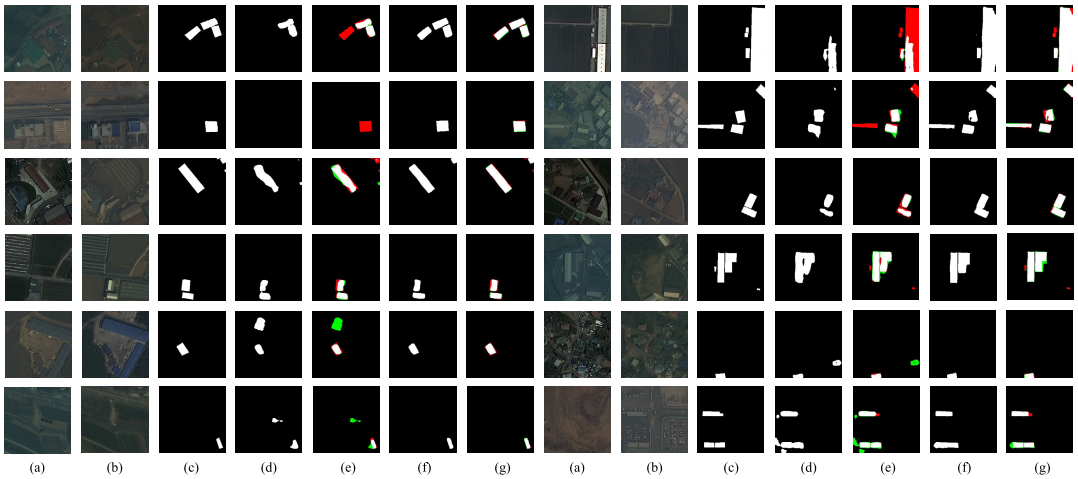


Fig. 9. Visualization of experimental results on the S2Looking dataset. (a) T1. (b) T2. (c) GT. (d) DSAMNet. (e) DSAMNet-diff. (f) GaMPF. (g) GaMPF-diff.

TABLE IV
OBJECTIVE DETECTION RESULTS OF THE ABLATION EXPERIMENTS

	baseline	w/o CE	w/o FsG	w/o DFM	GaMPF
CDD					
P	96.84	97.32	97.56	98.31	**98.75**
R	96.34	95.93	97.53	98.00	**98.33**
$F1$	96.59	96.62	97.55	98.16	**98.54**
OA	99.16	99.17	99.39	99.54	**99.64**
K	96.11	96.15	97.20	97.90	**98.33**
IoU	93.41	93.46	95.21	96.38	**97.12**
CLCD					
P	75.84	80.40	81.25	81.29	**84.06**
R	66.19	67.38	68.06	65.19	**68.36**
$F1$	70.69	73.31	74.07	72.36	**75.40**
OA	95.92	96.35	96.45	96.29	**96.68**
K	68.50	71.37	72.19	70.40	**73.64**
IoU	54.66	57.87	58.82	56.69	**60.52**
S2Looking					
P	72.16	73.84	73.41	**74.87**	73.46
R	25.02	35.36	42.80	37.53	**48.68**
$F1$	37.16	47.82	54.08	50.00	**58.56**
OA	98.98	99.07	99.13	99.09	**99.17**
K	36.76	47.40	53.67	49.59	**58.15**
IoU	22.82	31.42	37.06	33.33	**41.40**

TABLE V
COMPARISON OF FLOPS AND PARAMS OF CLCD BY EACH METHOD

Method	Params(M)	Flops(G)	F1	IoU
DTCDSCN	31.26	52.83	59.79	42.64
BiT	**3.49**	42.37	62.08	45.01
MFPNet	85.97	514.93	70.22	54.11
MSCANet	16.42	59.08	71.29	55.39
DSAMNet	33.39	**28.09**	71.24	55.33
SNUNet-CD	16.95	301.55	71.07	55.13
DR-TANet	12.04	219.33	<u>72.48</u>	<u>56.84</u>
GaMPF	69.64	60.21	**75.40**	**60.52**

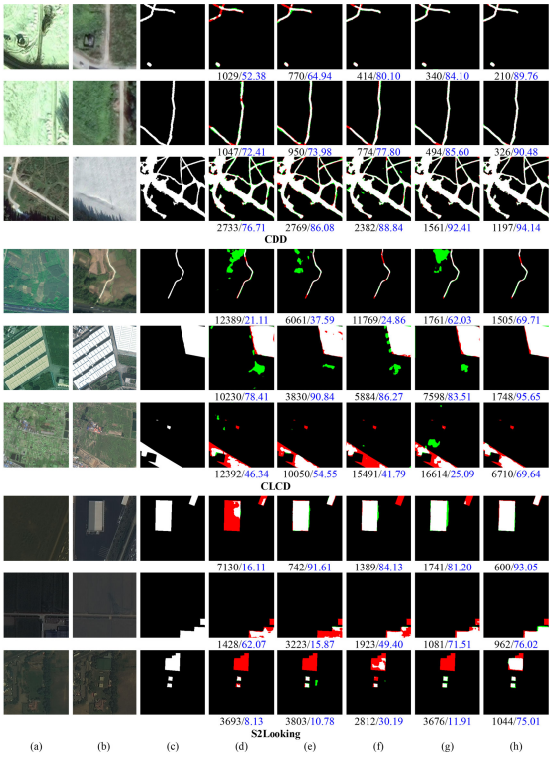


Fig. 10. Visualization of ablation study in the way of difference result between the prediction and GT. The total numbers of misclassified pixels are marked in black, and IoU values are marked in blue. (a) T1. (b) T2. (c) GT. (d) Baseline. (e) w/o CE. (f) w/o FsG. (g) w/o DFM. (h) GaMPF.

F. Efficiency Analysis of GaMPF

Table V analyzes the performance of each method from another perspective. The number of parameters (Params) and floating-point operations (Flops) are used to measure the complexity. It lists the Params and Flops of the methods involved in Section IV-C.

It can be observed that BiT is with the lowest Params, and DSAMNet is with the lowest Flops. The numbers of Params and Flops of DTCDSCN are higher than those of BiT, but the detection accuracy is lower instead. The computational load of MFPNet is extremely large, which is 8.55 times of GaMPF; however, the F1 and IoU metrics are lower by 5.18 and 6.41, respectively. The GaMPF has 69.64M Params and 60.21G Flops; meanwhile, it achieves the highest detection performance compared with other methods.

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V. Conclusion

In this article, a GaMPF based on CE for VHR RS image CD is proposed, which can make the change information contained in each feature map pass along within- and cross-temporal dual-path simultaneously. And the gated switches play the role of deciding whether and how much to send each position feature, and it aims at propagating real change information while filtering out invalid pseudo-change information. The gate switches are obtained by the cooperative estimation strategy, which can generate a key embedding representation for each feature map at the same scale simultaneously on a compact set of shared bases, making the difference features between bitemporal feature maps more significant. Finally, the DFM fuses the features at each scale multiple times in a concurrent short- and long-skip connection manner, so that the shallow pixel information and deep semantic information are reconciled into the same feature map to enhance the feature expression of the network. In addition, the effectiveness of the proposed method is proved by a large number of comparative experiments and data analysis, all of which outperform the existing SOTA methods.

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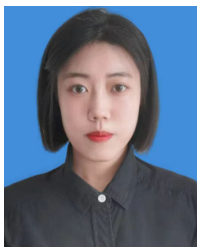
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